

2-40 Find the equivalent resistance between terminals A and B in Figure P2-40.

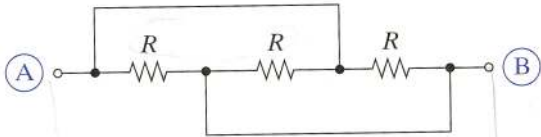


FIGURE P2-40

2–41 Use voltage division in Figure P2–41 to obtain an expression for v_L in terms of R , R_L , and v_S .

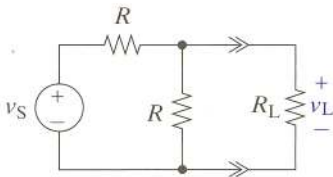


FIGURE P2–41

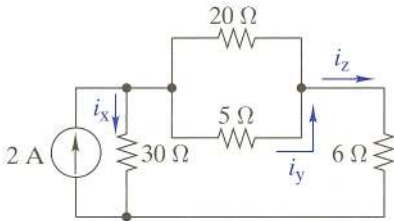


FIGURE P2-43

2-44 Find i_y and i_z in Figure P2-43.

2-48 **D** Select values for R_1 , R_2 , and R_3 in Figure P 2-48 so the voltage divider produces the two output voltages shown.

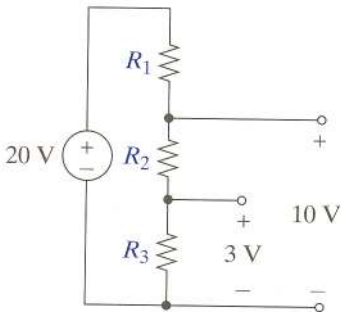


FIGURE P 2 - 4 8

2-54 Use circuit reduction to find v_x and i_x in Figure P2-54.

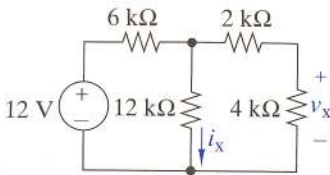


FIGURE P2-54

2-55 Use source transformation to find i_x in Figure P2-55.

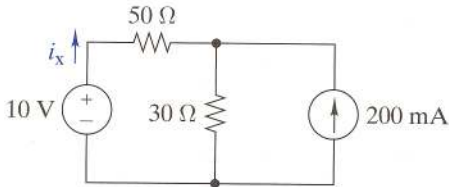


FIGURE P2-55

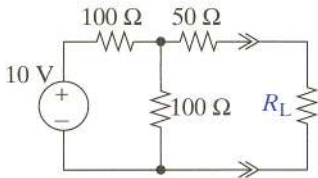


FIGURE P2-58

2-59 The voltage across R_L in Figure P2-58 is 2.5 V. Use source transformations to find R_L .

- 3-36 (a)** Find the Thévenin or Norton equivalent circuit seen by R_L in Figure P3-36.
- (b)** Use the equivalent circuit found in part (a) to find i_L in terms of i_S , R_1 , R_2 , and R_L .
- (c)** Check your answer in part (b) using current division.

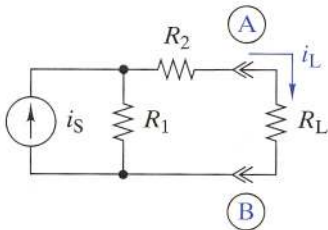


FIGURE P3-36

3-40 Find the Norton equivalent at terminals A and B in Figure P3-40.

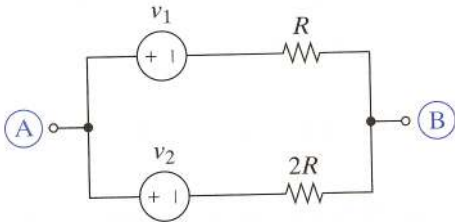


FIGURE P3-40

3-42 Figure P3-42 shows an active circuit with two accessible terminals. The output current is $i = 10$ mA when $v = 0$. The output voltage is $v = 6$ V when a 2.4 -k Ω resistor is connected between the terminals. How much current would this source deliver to a 6 -V battery?

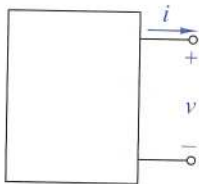


FIGURE P3-42